

ENG4200 — Coursework 2

Predicting customer churn: A comparative study of ML pipelines

Group ID: Telco-CW2-C

Group members: Ayush Bhandge (3148137B), Prathamesh Deepak Bagal (3154253B), Karthik Mandigiri (3155271M), Nathan Sidi Bakari (3071212S), Zefu Wang (3136524W), Jiangyue Yao Zhao (3084311Z)

Predicting customer churn

- Binary classification on the Telco Customer Churn dataset.
- Goal: build rigorous ML pipelines, not chase raw accuracy.
- Evaluate models along three axes:
 - predictive performance (F1, ROC-AUC),
 - computational efficiency (train + predict time),
 - overfitting resistance (CV gaps).
- Deliver: best pipeline + interpretable fallback.

Dataset characteristics

- 7043 samples × 20 features.
- Mixed feature types: charges, contract, internet service, tenure.
- Clear class imbalance: 26.5% churn.
- Meaningful heterogeneity → non-linear structures likely.
- Business cost of false negatives (missed churners) is high.

Dataset characteristics

Insert plot: *Churn class balance*

Standardised ML pipeline

- Identical structure for all models:
 1. Cleaning: TotalCharges coercion, collapsing "No internet/phone service".
 2. Preprocessing:
 - numeric → median impute + standardise
 - categorical → most-frequent + OHE (drop binary dummies)
 3. Classifier: LR / DT / SVM / RF / GB
- Ensures comparability and avoids data leakage.

Standardised ML pipeline

Insert diagram: *Pipeline schematic*

Five supervised learning models

- Logistic Regression (L2-reg, $C=0.01$)
- Decision Tree (depth=3)
- SVM (RBF, $C=0.1$)
- Random Forest (200 trees, depth=12)
- Gradient Boosting (200 estimators, $lr=0.05$)

These cover the entire supervised spectrum of the course materials: linear $\rightarrow$ kernel $\rightarrow$ tree $\rightarrow$ ensembles.

Accuracy & overfitting (CV)

- Key observations from 5-fold CV:
 - Random Forest: highest mean F1 (0.639), good ROC-AUC (0.845).
 - Logistic Regression: stable, minimal overfitting (gap = 0.01).
 - Decision Tree: competitive F1 but highest overfitting (gap = 0.05).
 - SVM & Gradient Boosting: strong ROC-AUC but not leading F1.

Accuracy & overfitting (CV)

Insert table: CV results

Final held-out performance

- On the independent test set:
 - Random Forest = best F1 (0.630)
 - Decision Tree = fastest (0.04 s)
 - SVM = highest cost (≥ 4 s)
 - Gradient Boosting = best ROC-AUC (0.846) but lowest F1

Final held-out performance

Insert table: Test benchmark (train time, predict time, F1, ROC–AUC)

Performance vs. training cost

- Using the frontier logic:
 - *A model is dominated if another is at least as fast and strictly better in F1.*
- Frontier contains only:
 - Decision Tree (fastest)
 - Random Forest (highest F1)
- Dominated: LR, SVM, GB (all slower + worse F1).

Performance vs. training cost

Insert plot: F1 vs train time (efficient frontier)

Justification of chosen pipeline

- Best F1 (0.630) and strong ROC-AUC.
- Only slightly slower than the Decision Tree (fractions of a second).
- Variance reduction via bagging + feature subsampling.
- Learns non-linear, interaction-rich decision rules without feature engineering.
- Diagnostics confirm good generalisation and no severe underfitting.

Random Forest diagnostics

- Confusion matrix: good recall + balanced precision.
- ROC curve: smooth, $AUC = 0.85$.
- Precision-recall curve: strong performance for minority class.
- Learning curve: training/validation F1 converge → moderate variance.
- Feature importances: Contract, tenure, charges, fibre optic.

Random Forest diagnostics

Insert plots:

- Confusion matrix
- ROC + PR curves
- Learning curve
- Top-k feature importances

Final model recommendation

- Primary model: Random Forest
 - best F1, strong ROC-AUC, low computational cost
 - robust variance behaviour
 - good diagnostics
- Secondary (interpretable) model: Decision Tree
 - extremely fast
 - transparent rules

Structured experiment design and methodological discipline were the key.

Thank you!

Questions?